

Exercise 3: Stored Procedures

Creating table Accounts:

```
CREATE TABLE Accounts (  
    AccountID NUMBER PRIMARY KEY,  
    CustomerID NUMBER,  
    AccountType VARCHAR2(20),  
    Balance NUMBER  
);  
  
SELECT * FROM Accounts;  
  
INSERT INTO Accounts VALUES (101,1,'Savings',10000);  
INSERT INTO Accounts VALUES (102,2,'Savings',5000);  
INSERT INTO Accounts VALUES (103,3,'Current',8000);  
  
COMMIT;
```

All rows fetched: 3 in 0.174 seconds

	ACCOUNTID	CUSTOMERID	ACCOUNTTYPE	BALANCE
1	101	1	Savings	10000
2	102	2	Savings	5000
3	103	3	Current	8000

Creating table employees

```
CREATE TABLE Employees (  
    EmployeeID NUMBER PRIMARY KEY,  
    EmployeeName VARCHAR2(50),  
    Department VARCHAR2(30),  
    Salary NUMBER(10,2)  
);  
  
INSERT INTO Employees VALUES (1,'Rahul','IT',50000);  
INSERT INTO Employees VALUES (2,'Priya','HR',45000);  
INSERT INTO Employees VALUES (3,'Rohan','IT',60000);
```

```

INSERT INTO Employees VALUES (4,'Anita','Finance',55000);

COMMIT;

SELECT * FROM EMPLOYEES;

```

Scenario - 1

```

CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest
AS
BEGIN
    UPDATE Accounts
    SET Balance = Balance + (Balance * 0.01)
    WHERE AccountType = 'Savings';

    COMMIT;
END;
/

EXEC ProcessMonthlyInterest;

```

OUTPUT

17 CREATE TABLE Employees (						
PROBLEMS	OUTPUT	DEBUG CONSOLE	TERMINAL	PORTS	QUERY RESULT	SCRIPT OUTPUT
All rows fetched: 4 in 0.160 seconds						
	ACCOUNTID	CUSTOMERID	ACCOUNTTYPE	BALANCE		
1	101	1	Savings	10100		
2	102	2	Savings	5050		
3	103	3	Current	8000		
4	104	4	Savings	20200		

Scenario - 2

```

CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus (
    dept IN VARCHAR2,
    bonusPercent IN NUMBER
)
AS
BEGIN

```

```

UPDATE Employees
SET Salary = Salary + (Salary * bonusPercent / 100)
WHERE Department = dept;

COMMIT;
END;
/

```

OUTPUT

All rows fetched: 4 in 0.152 seconds

	EMPLOYEEID	EMPLOYEEENAME	DEPARTMENT	SALARY
1	1	Rahul	IT	60500
2	2	Priya	HR	45000
3	3	Rohan	IT	72600
4	4	Anita	Finance	55000

Scenario - 3

```

EXEC UpdateEmployeeBonus('IT',10);

SELECT * FROM EMPLOYEES;

CREATE OR REPLACE PROCEDURE TransferFunds(
    sourceAcc IN NUMBER,
    destAcc IN NUMBER,
    amount IN NUMBER
)
AS
    balance NUMBER;
BEGIN
    SELECT Balance
    INTO balance
    FROM Accounts
    WHERE AccountID = sourceAcc;

    IF balance >= amount THEN
        UPDATE Accounts
        SET Balance = Balance - amount
    END IF;
END;

```

```

WHERE AccountID = sourceAcc;

UPDATE Accounts
SET Balance = Balance + amount
WHERE AccountID = destAcc;

COMMIT;

DBMS_OUTPUT.PUT_LINE('Transfer Successful');
ELSE
    DBMS_OUTPUT.PUT_LINE('Insufficient Balance');
END IF;
END;
/

```

OUTPUT

	PROBLEMS	OUTPUT	DEBUG CONSOLE	TERMINAL	PORTS	QUERY RESULT	SCRIPT
All rows fetched: 4 in 0.160 seconds							
		ACCOUNTID	CUSTOMERID	ACCOUNTTYPE	BALANCE		
1		101	1	Savings	6100		
2		102	2	Savings	9050		
3		103	3	Current	8000		
4		104	4	Savings	20200		